Aegora Isle

Simulation Charts Reference Guide

Plain-language explainer of what each test-run chart shows

This document explains, in everyday language, what each chart from our test runs is showing — and exactly what settings (how many rounds, which conditions, how many characters) were used to produce it. No technical or programming background is needed to read this.

Companion to the more technical *Governance & KPI Reference Guide*. Aegora Isle project — July 2026.

Before the Charts: The Basic Setup

Aegora Isle is a computer simulation of a tiny made-up island economy. Twenty pretend "characters" live there: some dig up a raw material from a shared pool, some turn that material into finished goods, some trade those goods, and — depending on the setup — some act as a kind of "government" trying to keep things fair and sustainable. We run this simulation over and over under different rules and different conditions, and every chart in this document is one of those test runs, turned into a picture so we can see what happened at a glance.

The three "government styles" we compare

- **No government (called G1):** nobody manages anything. Every character just does whatever they want, with no limits and no consequences.
- **Fixed rules (called G2):** a simple rulebook that never changes — a cap on how much material anyone can dig up per round, an occasional fine if someone goes over that cap, and a scheduled hand-out of money from richer characters to poorer ones every so often.
- **A learning government (called G3):** instead of fixed rules, this is a computer program that watches what's happening on the island and gradually adjusts its own rules over time — its cap, its hand-out amount, how strict it is — based on trial and error, the way a person might get better at a game the more they play it.

The four "weather conditions" we test

We test every government style under four different starting conditions, mixing "how much material is available" with "how often bad luck strikes":

- **Scenario A:** Plenty of material, bad luck is rare. The easy case.
- **Scenario B:** Plenty of material, but bad luck strikes often. Tests resilience under repeated shocks.
- **Scenario C:** Little material to begin with, bad luck is rare. A slow, steady squeeze.
- **Scenario D:** Little material, and bad luck strikes often. The hardest case — our main stress test.

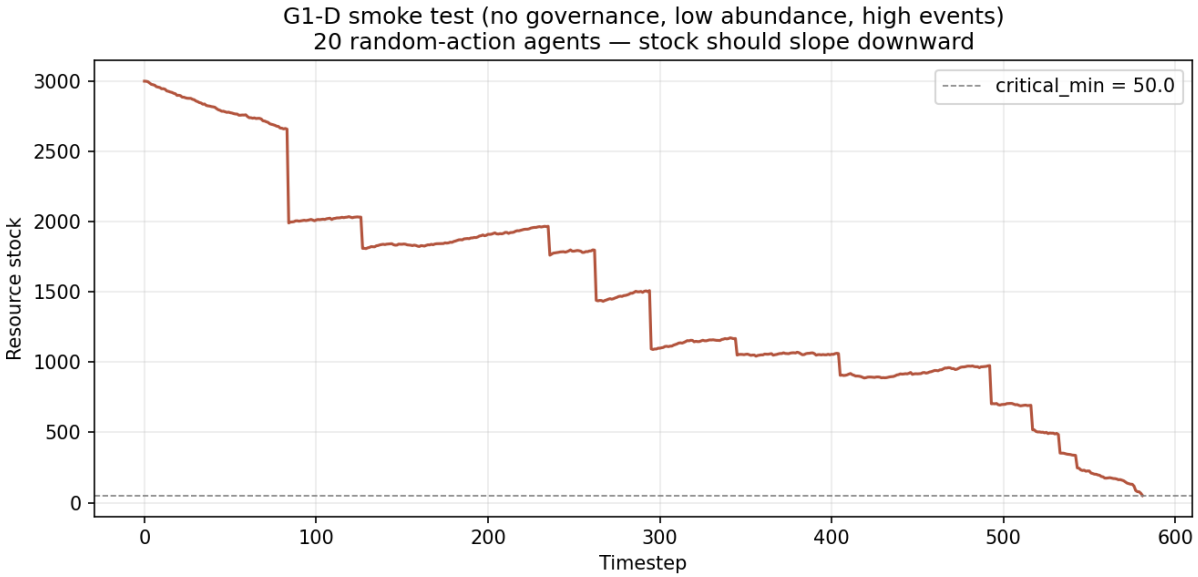
What a "round" (or "cycle") means

One round is one full turn where every single character on the island gets to act exactly once. We let the simulation run for many rounds at a time — sometimes a few hundred, sometimes a few thousand — and every chart below states exactly how many rounds were used to make it.

Why we always start from the same "dice roll"

To make comparisons fair, we always start each simulation from the exact same random starting point — think of it like always shuffling a deck of cards the same way before dealing. That way, if two government styles turn out differently, it's because of the RULES being different, not because one run got luckier than the other.

Chart 1 — First Health Check: Does the Material Run Out?

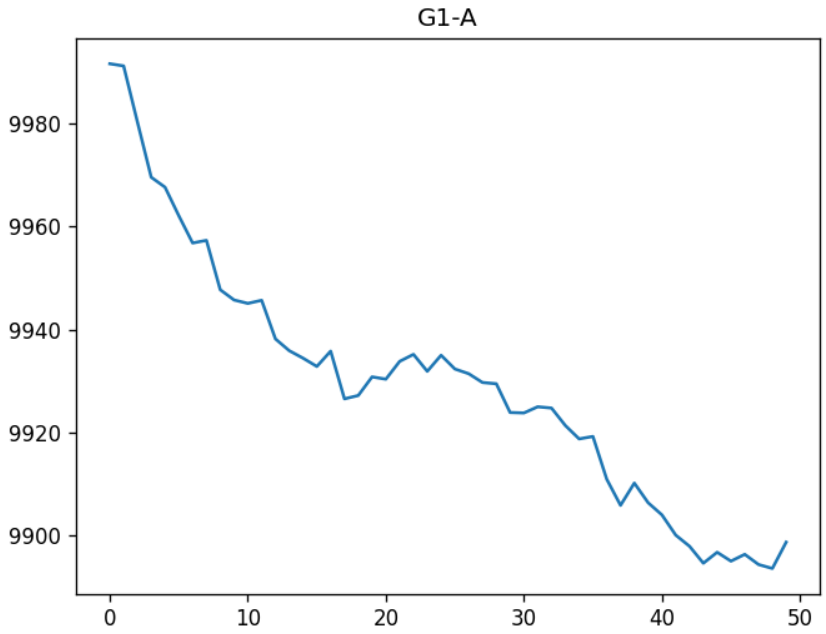


Government style:	No government (G1) only
Weather condition:	Scenario D — little material, frequent bad luck (the hardest case)
Number of characters:	20
Rounds:	Allowed up to 1,000 — actually stopped at round 581, when the material ran out
How characters behaved:	Completely randomly — no strategy, no learning

What this shows, in plain terms: this was our very first sanity check. It asks one simple question: if nobody is managing the shared material and conditions are harsh, does the material actually run out over time — the way common sense says it should? The line trends steadily downward and eventually hits bottom, confirming: yes, with no oversight and tough conditions, the shared resource does get used up.

What to notice: the line isn't perfectly smooth — it has sudden sharp drops. Those are the random "bad luck" events (natural disruptions) hitting the shared material pool.

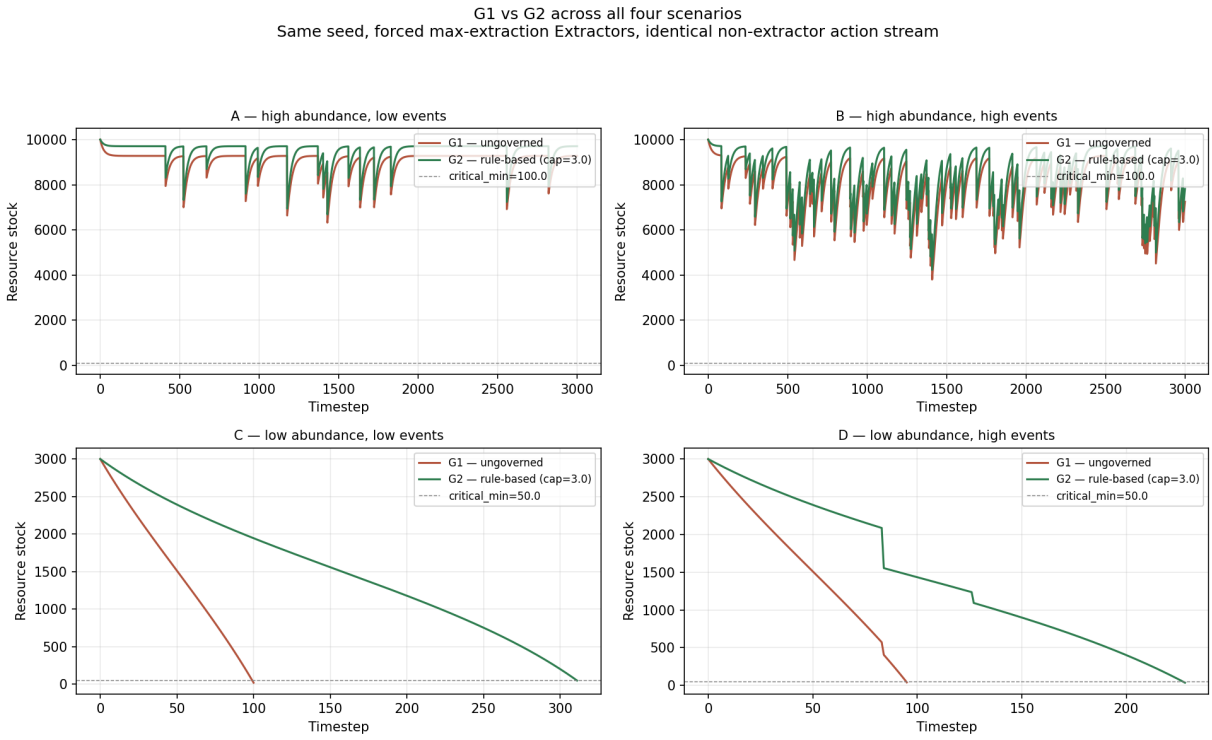
Chart 2 — Same Check, Under Easier Conditions



Government style:	No government (G1) only
Weather condition:	Scenario A — plenty of material, bad luck is rare (the easiest case)
Number of characters:	20
Rounds:	Up to 1,000
How characters behaved:	Completely randomly — no strategy, no learning

What this shows: the same check as Chart 1, but under gentle conditions instead of harsh ones. Compare the two: under easy conditions the material barely gets dented, whereas under harsh conditions (Chart 1) it's completely gone. Seeing the simulation react sensibly to different starting conditions — not just always collapsing no matter what — is exactly what we want to confirm before trusting any of the later, more complex charts.

Chart 3 — Comparing "No Rules" vs "Fixed Rules"



Government styles:	No government (G1) vs. Fixed rules (G2), side by side
Weather conditions:	All four (A, B, C, and D) shown as four small charts
Number of characters:	20
Rounds:	Up to 3,000 per weather condition
How characters behaved:	A deliberate worst-case test: every material-digging character was forced to always ask for the maximum possible amount, every single round — not normal behaviour, but a stress test designed to show the rules' effect clearly, without random luck getting in the way

What this shows: in every one of the four weather conditions, the "Fixed rules" line lasts noticeably longer than the "No government" line before the material runs out. That's the fixed cap on digging doing its job — it doesn't prevent the material running out entirely under this extreme stress test, but it clearly slows the process down.

Note: an earlier, single-condition version of this same check exists in our files (covering only the hardest weather condition) — this chart is the expanded version covering all four conditions at once, and supersedes it.

Chart 4 — All Three Government Styles, Quick Summary



Government styles:	All three — No government, Fixed rules, and the Learning government
Weather conditions:	All four (A, B, C, and D)
Number of characters:	20
Rounds:	300 per weather condition
How characters behaved:	For the first time, characters were genuinely learning from experience as they went, using a simplified trial-and-improve technique — not random, not forced to the maximum

What this shows: three side-by-side bar charts comparing fairness (how evenly wealth is spread), how fast the material is running out, and how much material is left — across all three government styles and all four weather conditions.

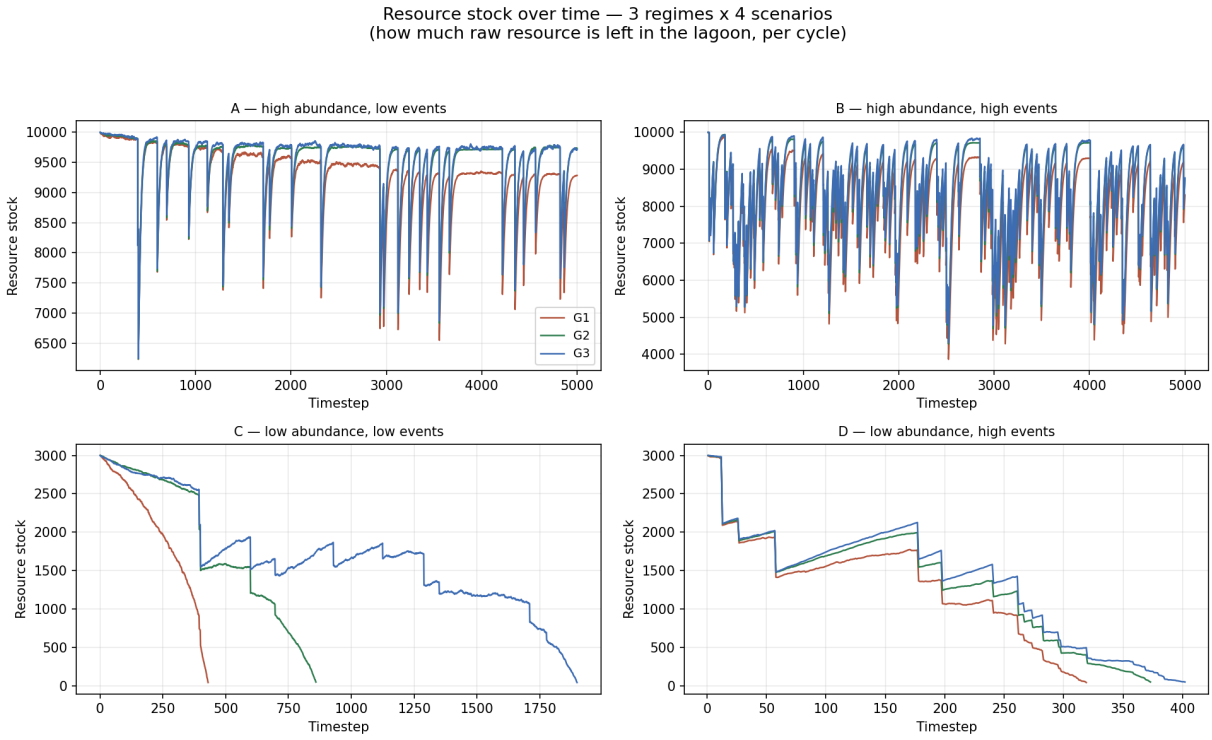
Something worth flagging honestly: the Learning government's numbers in this particular chart should *not* be read as "the learning government performed worse." It simply hadn't had enough practice rounds yet to learn anything — it was still behaving close to randomly. This is exactly why we ran the longer version shown next.

Charts 5–9 — A Longer, More Detailed Look

The five charts on the following pages all come from the **same, longer test run** — we simply look at five different aspects of it. This run was made deliberately long enough for the Learning government to actually cross the point where it's allowed to start learning from what it's seen (it first needs to gather 1,000 rounds worth of experience before it's allowed to begin adjusting itself).

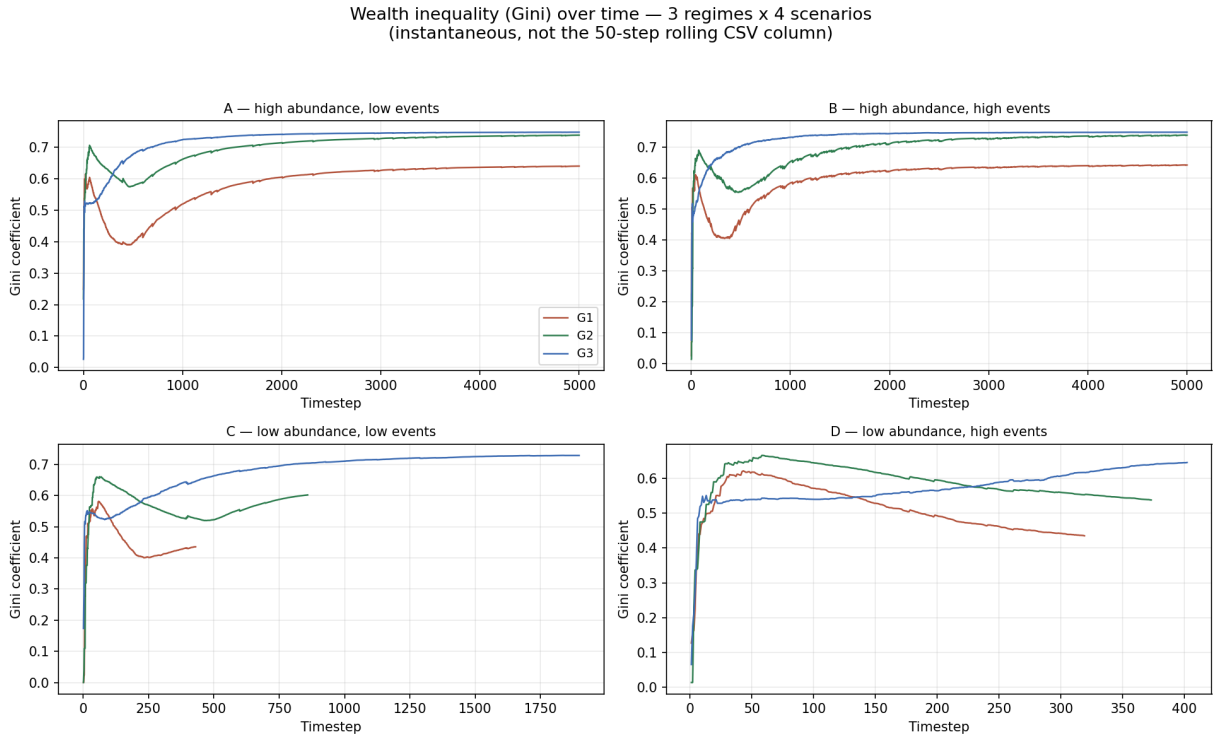
Government styles:	All three — No government, Fixed rules, and the Learning government
Weather conditions:	All four (A, B, C, and D)
Number of characters:	20
Rounds:	5,000 per weather condition (some runs ended earlier if the material fully ran out first)
How characters behaved:	Genuinely learning from experience as they went, same as Chart 4, just given much more practice time

Chart 5 — How Much Material Is Left, Over Time



Four small charts, one per weather condition, each showing how much material remains in the shared pool as time passes, with one line per government style. Sharp drops are bad-luck events hitting the pool; the climbs after each drop are the material naturally recovering. In the "plenty of material" conditions (A and B), the island holds fairly steady. In the "little material" conditions (C and D), all three government styles eventually run out — but the "No government" line runs out noticeably fastest.

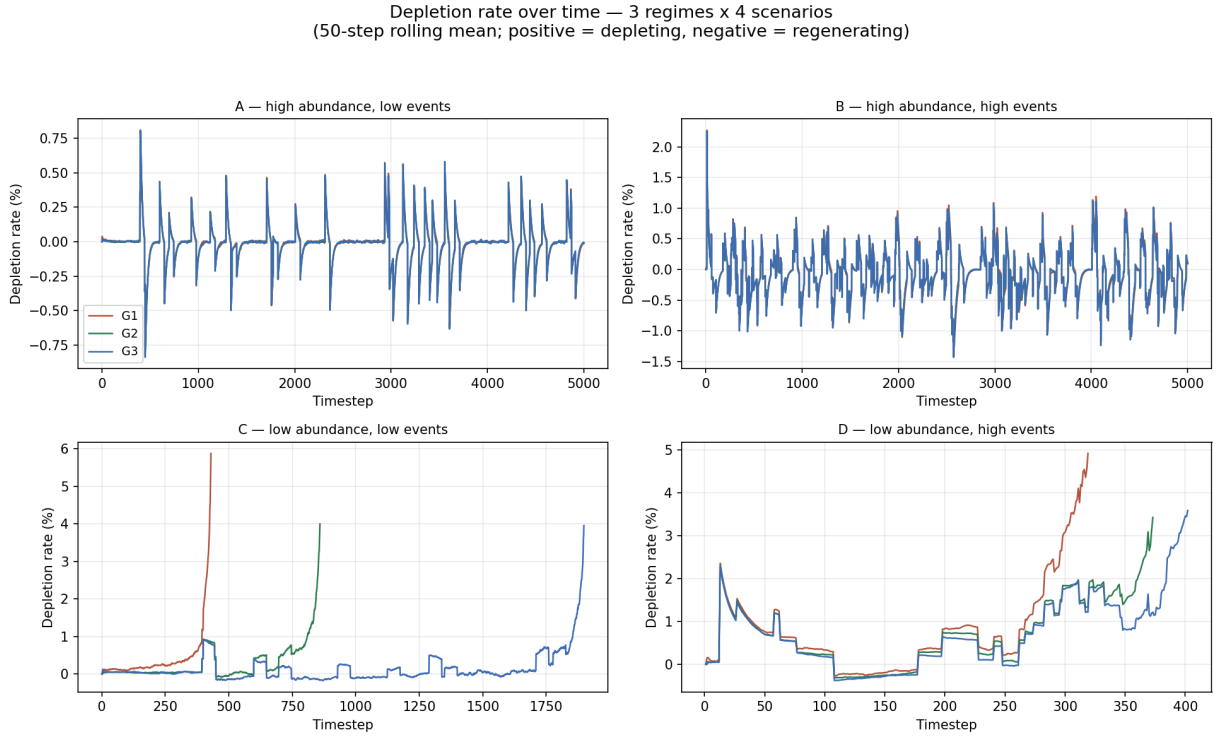
Chart 6 — Fairness Over Time



This tracks how unevenly wealth is spread across the 20 characters, as time passes. A low number means everyone has roughly the same amount; a high number means a few characters have almost everything while the rest have very little.

An honest, interesting result: across every single weather condition, the Learning government's line settles at the *highest* (least fair) level — not the lowest, which is what we'd eventually hope to see from a well-trained learning government. As explained on the previous page, this is because it still hadn't had enough practice to learn a sensible strategy — it was largely still making close to random decisions throughout this test.

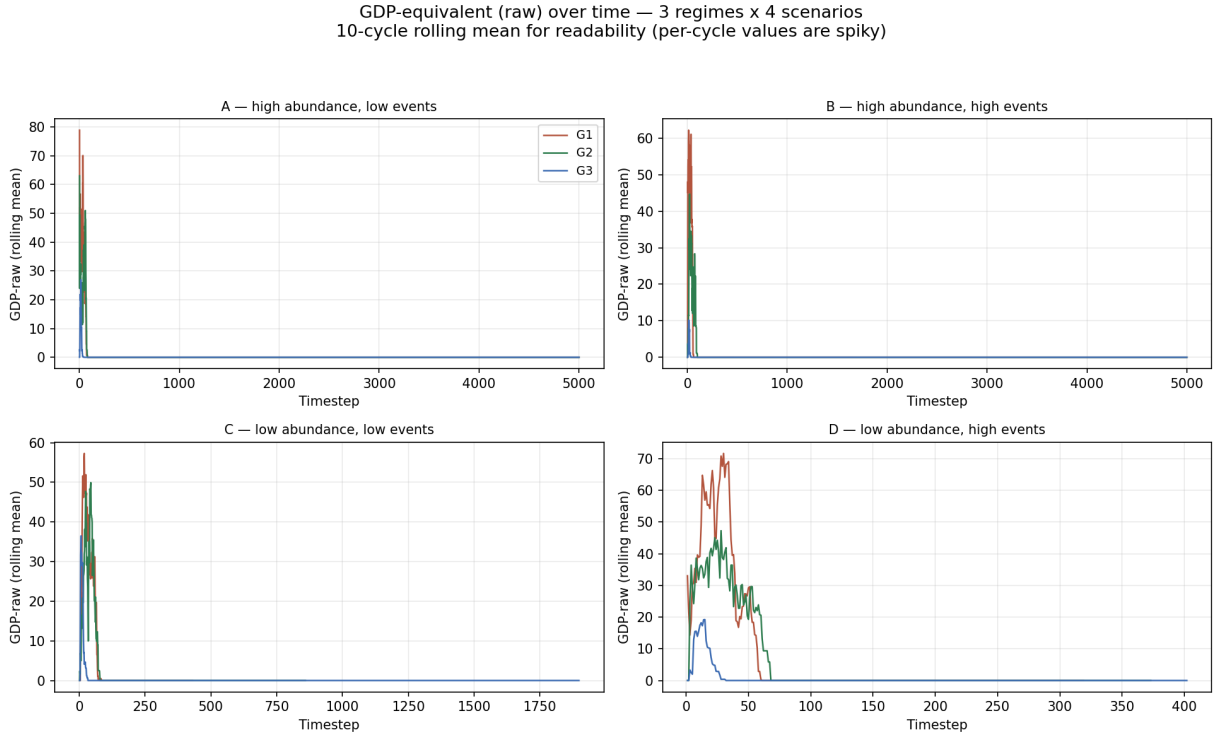
Chart 7 — How Fast the Material Is Changing



Rather than showing the total amount of material left, this shows how quickly it's being used up (or replenished) at each moment. Spikes upward mean a bad-luck event just hit; dips below zero mean the pool is recovering faster than it's being drained.

What to notice: right before the material fully runs out (visible in the low-material weather conditions), this number shoots up sharply just beforehand — a kind of early-warning sign, visible just ahead of a full collapse.

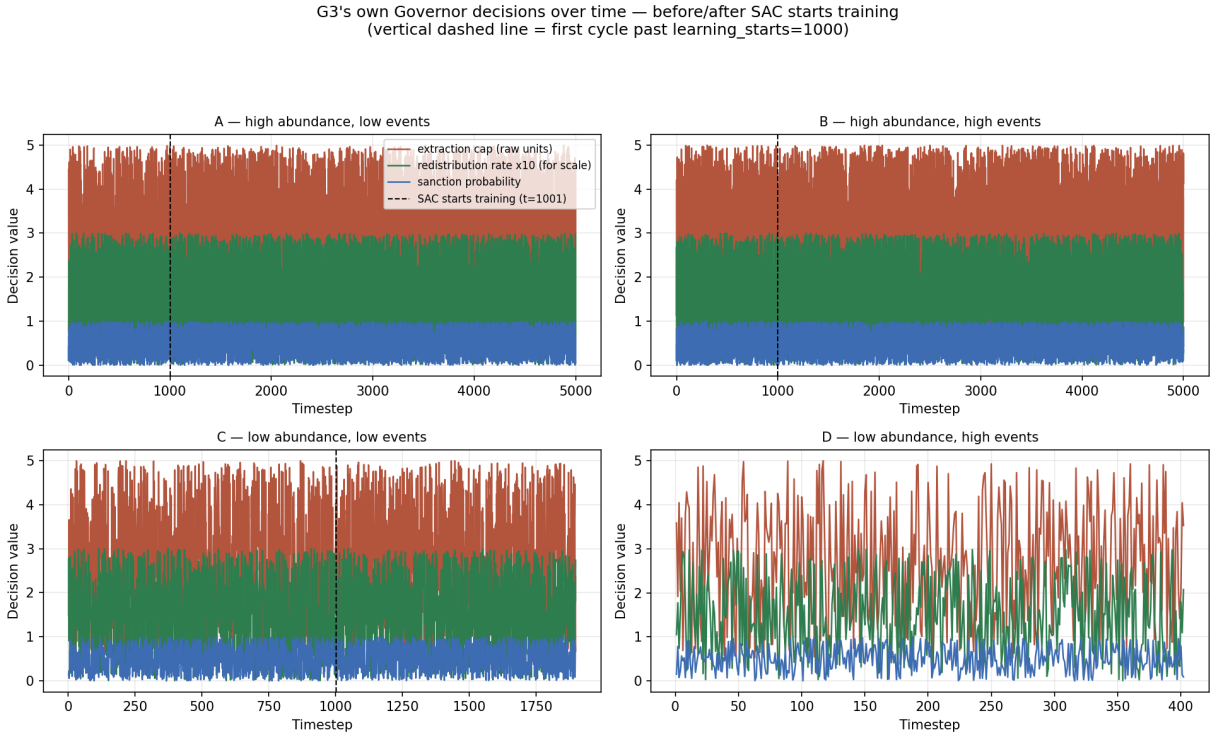
Chart 8 — Economic Activity Over Time



This is a rough stand-in for "how much economic activity — trading and making goods — is happening" at any given moment.

An unexpected, genuinely useful finding: across every single weather condition and every government style, there's a burst of activity in the very first 50 to 100 rounds, and then it drops to almost nothing for the rest of the run — thousands of rounds with barely any trading happening at all. Our working guess is that characters start with only a small amount of money, spend it quickly in that opening burst, and then have no reliable way to earn more, so trading grinds to a halt. This is a good example of exactly what these check-in charts are for: catching a real pattern worth digging into further, rather than confirming something we already expected.

Chart 9 — What the Learning Government Was Deciding



This last chart is specific to the Learning government. It tracks the three "dials" it controls over time: how much material-digging it allows, how much wealth redistribution it schedules, and how likely it is to fine rule-breakers. The dashed vertical line marks the exact round where it becomes *allowed* to start learning — after collecting 1,000 rounds worth of experience.

What to notice: the pattern doesn't visibly change right at that dashed line — it still looks just as random after it as before. We can confirm from the underlying numbers that learning had technically begun by that point, but the roughly 4,000 remaining rounds simply wasn't enough practice for a visible change in behaviour to show up yet. Learning systems like this one typically need far more practice than we gave it here before they start looking noticeably different from random guessing.

Important Things to Keep in Mind

- Every chart in this document comes from a short test run, not the full, final scientific study.
- The "Fixed rules" and "Learning government" settings used throughout are first-draft numbers, not yet fine-tuned.
- The Learning government especially needs much more practice time before its results can be trusted to mean anything.
- None of these charts should be treated as a final finding. They are "does the simulation basically behave the way we expect" checks — which is exactly what they were built for.